
Backbone-Modular Neuro-Symbolic Post-Correction for OCT Denoising via Explicit Predicate Maps and Stable Fuzzy Allocation

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ABSTRACT Denoisers for optical coherence tomography are trained to reduce a pixel error, so they can raise fidelity while lowering what a clinician reads, the contrast between retinal layers and the sharpness of their boundaries. We repair those properties afterwards, leaving the denoiser untouched. A small wrapper on the frozen denoiser measures six named clinical properties from the noisy input and its output alone, needing no clean image at test time, turns them into a per-pixel decision through five rules in fuzzy logic, bounds how far the image may move, and keeps a proposed change only while stated constraints hold, so every change is attributable to a named property at a named place. We prove that the decision cannot jump, by an exact rather than a loose constant, and give conditions under which the correction cannot lower contrast. One set of rules, budgets and weights serves four denoisers, only a small head being refitted. On all four, contrast rises by 3.5 to 11.3 percent and speckle suppression by 23.7 to 57.1 percent, for 0.17 to 0.98 dB of fidelity, and 21 of 24 gains clear the spread across seeds. Carried unchanged to two further datasets, 40 of 48 pairs improve.

INDEX TERMS Clinical quality assessment, fuzzy logic, image denoising, neuro-symbolic reasoning, optical coherence tomography.

I. Introduction

OPTICAL coherence tomography (OCT) is the standard non invasive way to image the retina [1]. Every OCT image carries speckle, a multiplicative disturbance that arises from coherent light scattering inside tissue [2]. Speckle blurs the border between retinal layers, lowers the contrast between tissue and background, and makes automatic layer segmentation less reliable [3].

Deep denoisers remove much of that disturbance, but they are almost always trained to minimise the squared error against a clean reference, so a denoiser can raise the peak signal to noise ratio (PSNR) of a scan while making it less useful to a clinician. Layer boundaries lose definition, the contrast between tissue and the dark vitreous falls, and thin structures with weak signal fade into the background. None of these losses is visible in a PSNR number.

A. Motivation

The gap we address is not a missing denoiser. It is that nothing sits between the denoiser and the clinical

reader to state what a good OCT image should look like and to repair the specific ways a denoised image falls short. Another neural block would hide the reason for every change inside learned weights, while a clinician, a regulator or a reviewer who sees a modified scan needs to know which property was judged deficient, where, and how large a change was allowed. This paper therefore treats post-correction as its own problem. We keep the denoiser frozen and attach a small wrapper, written as named image properties, explicit logical rules over them, and explicit limits on the size of any change, that decides where the image should change and by how much. Every learned map in the decision path is bounded, and its effect reaches the image only through a named rule.

B. New ideas in this paper

The new ideas are four. Edge filters, gating maps and fuzzy operators are not among them.

- 1) Six clinical properties that need no clean image. Section V-A defines six properties computed from

the noisy input and the denoiser output alone, so they are available on any scanner at test time. Each returns a score for the whole image and a map of where it fails, both as formulas.

- 2) A rule layer whose decision cannot jump. Section V-B builds the allocation from five rules joined by a conjunction with bounded slope, Section VII proves that a shift in the evidence moves the decision by at most a constant that the rule layer itself fixes, and Section IX-G measures the response the images actually produce against it.
- 3) A safety layer that decides how much of a proposal to keep. Section VI-B states three constraints at three scales, and how many of them hold sets the fraction of the proposed change that is applied. A margin condition under which the correction cannot lower the contrast to noise ratio is proved and then tested on every test image.
- 4) One design for four denoisers. Only the head that reads backbone features changes with the backbone. The rules, the budgets, the thresholds and the loss weights do not, the shared wrapper is a small fraction of the denoiser it is attached to, and Section IX-C carries it to two other datasets with no adaptation at all.

II. Related Work

A. Denoising of OCT images

Learned denoisers for OCT fall into three families. Convolutional networks such as DnCNN [4] learn the noise as a residual and subtract it. Designs that gate or reweight channels rather than pass them through a nonlinearity, such as NAFNet [5] and KBNNet [9], control what each contributes. Transformers such as SwinIR [6] use windowed self attention. Classical estimators remain in use, including non local means [10] and wavelet coefficient weighting across repeated acquisitions [11], and self supervised training removes the need for clean targets [7]. Methods built for OCT address the scarcity of paired data directly, through triplet cross fusion [12] and through sparse representation [14], although the earlier of those still needs one high quality scan of the same eye [13].

All of these methods are trained to reduce a pixel error, so the quantity they optimise is not the quantity a clinician reads. Work on quality assessment for OCT, whether a learned classifier [15], a computed index [20] or the OSCAR-IB expert criteria [21], measures clinical quality but does not act on it. The present paper closes that loop by turning measured quality into a correction applied under stated limits.

B. Confidence and uncertainty in medical imaging

Dropout as an approximate Bayesian posterior [16], task weighting driven by learned variance [17] and

deep ensembles [18] are the standard ways to attach a confidence to a prediction. In OCT the networks that segment layers [3] or judge scan quality [15] report an accuracy rather than an uncertainty, so a reader learns how often the model is right and never where it is unsure. In almost all of that work the value is reported and then left alone. We instead feed such a map into the correction decision, and Section IX-F reports that the map we built is not a confidence in this sense.

C. Symbolic reasoning and expert networks joined with neural networks

Neuro symbolic computing [19] couples a learned model with explicit rules, and differentiable fuzzy logic makes the coupling trainable because the connectives become smooth functions of continuous truth values [8]. In medical imaging the pattern has generated spinal reports by pairing a generative network with a symbolic graph of domain knowledge [27], predicted patient deterioration with a probabilistic logic program over convolutional features [28], and interpreted corneal topography through a knowledge graph attached to a learned feature model [34]. Fuzzy membership functions have been embedded in networks that process OCT directly, where a rough fuzzy discretisation guides the segmentation of fluid lesions [29], and pixel level fuzzy rules have replaced learned feature extraction, so what the model attends to reads as a logical statement [30].

A parallel line builds an image model from cooperating experts. Steered mixtures of experts combine constant experts through soft gating kernels with an explicit steering term [31], also for OCT speckle removal [32], and recent restoration networks route an input through experts of different capacity according to its difficulty [33]. Outside imaging, evolving fuzzy models identify actuator dynamics [24], learned models are joined to a physical forward operator in ultrasound tomography [25], and sensitivity to initial conditions is treated as a source of uncertainty in simulation [26]. All rest on the principle we rely on, that a learned component is easier to trust when part of the decision is written in a language a person can read. What is missing is a rule layer that acts on the output of a strong denoiser, names the clinical properties it protects, and limits how far it may move the image, which is the object studied here.

III. Notation and Problem Statement

A. Notation

Images are real valued arrays of height H and width W with values in $[0, 1]$. We write $\mathbf{y}$ for the noisy B-scan that comes off the scanner and $\mathbf{x}$ for the clean reference obtained by averaging many repeated scans of the same location. A pixel index is written p , so y_p is one pixel of $\mathbf{y}$.

TABLE 1. Symbols used in the paper, in order of first use.

Symbol	Meaning
$\mathbf{y}$	Noisy input B-scan
$\mathbf{x}$	Clean reference B-scan
f_θ	Frozen denoising backbone with parameters θ
$\mathbf{b}$	Backbone output, $\mathbf{b} = f_\theta(\mathbf{y})$
ϕ	Trainable parameters of the wrapper
$\mathbf{c}$	Cooperation map with values in $[0, 1]$
$\mathbf{u}$	Demand map, $\mathbf{u} = \mathbf{1} - \mathbf{c}$
s_i	Score of clinical property i , a number in $[0, 1]$
$\mathbf{m}_i$	Failure map of clinical property i , values in $[0, 1]$
τ_i	Decision threshold for property i
Δ	Raw multiplicative gain proposed by the gain network
λ	Per pixel strength map with values in $[0, 1]$
$\mathbf{a}$	Allocation map produced by the rule layer
α	Gain after allocation, strength and clamping
$\mathbf{g}$	Background gate that suppresses change in the background
$\mathbf{v}$	Gated multiplicative change added to the backbone output
δ_e	Additive edge correction
$\mathbf{q}$	Candidate image before the safety decision
$\hat{\mathbf{x}}$	Final output of the wrapper
$\mathcal{T}, \mathcal{B}$	Sets of tissue pixels and background pixels
$\mu_{\mathcal{T}}, \mu_{\mathcal{B}}$	Mean value over tissue and over background
$\sigma_{\mathcal{B}}$	Standard deviation over background

The symbol $\odot$ is the Hadamard product, $(\mathbf{t} \odot \mathbf{t}')_p = t_p t'_p$. Table 1 lists the symbols that recur across sections, and a symbol used in one place only is defined where it is introduced.

B. The optimisation problem

The denoiser is given and is not changed. The task is to produce an image that beats $\mathbf{b} = f_\theta(\mathbf{y})$ on named clinical measures, stays close to it everywhere, and can explain every change it makes. The only free quantities are the wrapper parameters ϕ , namely the cooperation head, the gain network, the edge network, the strength predictor and the rule layer, while the backbone parameters θ are held fixed throughout. Let $\mathcal{D}$ be a set of paired scans and $\hat{\mathbf{x}}(\phi)$ the output of the wrapper. We minimise the expected value of a weighted sum of six terms with fixed weights w_c, w_p, w_o, w_s and w_r ,

$$\min_{\phi} \mathbb{E}_{(\mathbf{y}, \mathbf{x}) \sim \mathcal{D}} \left[\mathcal{L}_{\text{fid}} + w_c \mathcal{L}_{\text{clin}} + w_p \mathcal{L}_{\text{pred}} + w_o \mathcal{L}_{\text{coop}} + w_s \mathcal{L}_{\text{edge}} + w_r \mathcal{L}_{\text{reg}} \right], \quad (1)$$

where $\mathcal{L}_{\text{fid}}$ keeps the output near the clean reference, $\mathcal{L}_{\text{clin}}$ rewards improvement of the clinical measures over the backbone, $\mathcal{L}_{\text{pred}}$ asks four of the property scores to rise above the value the backbone already reaches, $\mathcal{L}_{\text{coop}}$ ties the amount of correction to the estimated correction demand, $\mathcal{L}_{\text{edge}}$ supervises the gradient field, and $\mathcal{L}_{\text{reg}}$ collects the bounds on how large and how rough the change may be together with four terms that protect the background. Section VIII gives every term in full with its weight. Three constraints are checked on every image, stated formally in Section VI-B. The aggregate clinical quality should not fall by more than a fixed tolerance, no single property should fall by more than a fixed tolerance while another rises, and the relative change from the

backbone output should stay below a fixed bound. How many hold decides how much of the change is kept, so the rule is a majority test and not three hard guarantees. The objective is differentiable in ϕ , including the rule layer, because every logical connective is a piecewise linear surrogate, so it is solved with AdamW [23]. During training the constraints enter as penalties. At test time they are evaluated exactly on each image and the number satisfied sets the blend weight through the majority rule.

IV. Frozen Backbone and Bounded Proposals

A. Overview

Fig. 1 shows the whole method as six stages, taken in order here and in the two sections that follow, two to a section. Stage one asks where the backbone output can be left alone, stage two what change is available, stage three what is wrong and where, stage four where the change should be applied, stage five how, and stage six whether it is allowed at all.

B. Stage one, the backbone and the cooperation map

The backbone is frozen after training and gives $\mathbf{b} = f_\theta(\mathbf{y})$. The cooperation map $\mathbf{c}$ says where $\mathbf{b}$ can be left alone and where a change is wanted. A two layer network h_ϕ reads a feature tensor $\mathbf{z}$ from inside the backbone and produces it,

$$\mathbf{c} = \sigma \left(\frac{h_\phi(\mathbf{z}) + \beta_0}{\tau_0} \right), \quad (2)$$

where σ is the logistic function, $\tau_0 > 0$ is a learnable temperature and β_0 a learnable offset. The complementary demand map is $\mathbf{u} = \mathbf{1} - \mathbf{c}$.

Only this head depends on the backbone, because architectures expose feature tensors with different channel counts. We read the first encoder stage of NAFNet and KBNNet, the first convolution of DnCNN and the shallow feature convolution of SwinIR, and the head is a pair of convolutions sized to whichever tensor it is given, the widths being the only thing that differs and listed in the supplementary file. Everything after this stage is shared, which is the precise sense in which the method is modular rather than architecture free.

C. Stage two, the two correctors

Two changes of deliberately different kinds are proposed.

The gain network proposes a multiplicative change. It runs at quarter resolution and emits a brightness channel passed through a rectifier, so it can only brighten, and a contrast channel passed through a hyperbolic tangent, so it can go either way. Their sum is scaled by $\alpha_{\text{max}} = 0.15$ to give the raw gain Δ . A multiplicative form suits speckle, which is itself multiplicative [2], so bright tissue receives a larger absolute change than dark background for the same relative adjustment.

The edge network proposes an additive change δ_e from four channels, namely the intensity normalised

residual, the edge magnitude of the backbone output, the backbone output, and an unsharp mask. Its output is bounded to ± 0.08 and scaled by a learned factor. An additive form suits boundaries, which should be sharpened by the same amount in a bright layer as in a dark one.

A third small network predicts a per-pixel strength map λ with values in $[0, 1]$ from the failure maps of Stage three. It modulates how much of the raw gain survives at each pixel.

V. Clinical Properties and Fuzzy Allocation

A. Stage three, the six clinical properties

Each of the six properties names a way a denoised OCT image is known to fail and each is written as a formula so it can be recomputed by anyone. Table 2 gives, for every property, the reason it exists, its score $s_i \in [0, 1]$ as a weighted sum of named subterms, its map $\mathbf{m}_i \in [0, 1]^{H \times W}$ in closed form, where that map is large, and the threshold τ_i that separates satisfied from unsatisfied, with every subterm defined beneath the table. A larger score means the property is better satisfied and a larger map value means stronger local evidence that it is not. None of these quantities needs a clean reference. They are computed from the noisy input and the backbone output only, so they remain available at test time on a scanner the method has never seen.

B. Stage four, the rule layer

The rule layer decides where the gain proposal should act through five named rules, each casting a signed vote.

Each property score is first turned into a soft failure signal by comparing it with its threshold,

$$f_i = \sigma(\kappa(\tau_i - s_i)), \quad \kappa = 10, \quad (3)$$

so f_i is near one when property i is failing and near zero when it is satisfied. The same soft comparison is used to test the cooperation map and the strength of a proposal against their own thresholds. Each corrector produces that strength as a per-pixel map from the backbone output and the failure maps it serves, large where its property is far below threshold. Before the rules fire, that strength is multiplied by one half plus the demand map, so the same proposal looks stronger to the rules at a pixel where the head reports low cooperation. Conjunction uses the Lukasiewicz form $\text{AND}(a, b) = \max(0, a + b - 1)$, which is piecewise linear and has a bounded slope, which is what makes the bound in Section VII tight.

The five rules are the following. Rule one fires when the cooperation map is high and the proposal is weak, and it votes against changing the image. Rule two fires when the map is low and the proposal is strong, and it votes for changing it. Rule three fires when a relevant property is failing, and it votes for changing it. Rule four fires when the map and the proposal strength are both

middling, and it votes weakly for changing it. Rule five fires when several correctors want the same pixel, and it votes against changing it. Write r_1 to r_5 for the resulting truth values.

The allocation at each pixel is the bounded weighted vote

$$a = \underbrace{B_{\min} + (B_{\max} - B_{\min}) \sigma(\beta)}_{\text{base, no rule firing}} - C_1 w_1 r_1 + C_2 w_2 r_2 + C_3 w_3 r_3 + C_4 w_4 r_4 - C_5 w_5 r_5, \quad (4)$$

where β is a learnable base logit, $w_k = \sigma(\theta_k) \in (0, 1)$ are learnable rule weights and the budgets are $B_{\min} = 0.25$, $B_{\max} = 0.50$, $C_1 = 0.15$, $C_2 = 0.25$, $C_3 = 0.15$, $C_4 = 0.10$ and $C_5 = 0.10$.

The budgets satisfy $B_{\max} + C_2 + C_3 + C_4 = 1$ and $B_{\min} - C_1 - C_5 = 0$, so since every truth value lies in $[0, 1]$ and every weight in $(0, 1)$, the value of Equation (4) lies in $[0, 1]$ for every input and every parameter value and no clipping is ever needed. Without the identity a clamp is needed, the allocation saturates at one end of it, as it did on every pixel in the submitted version of this work, the rule weights receive no gradient, and the stability result of Section VII is vacuous.

VI. Bounded Change and the Safety Decision

A. Stage five, applying the change

The admitted gain is the raw proposal modulated by the allocation and the strength map, then bounded,

$$\alpha = \text{clamp}(\Delta \odot \mathbf{a} \odot \lambda, -0.20, 0.20). \quad (5)$$

A gate suppresses any change in the dark background,

$$\mathbf{g} = \sigma(20(\mathbf{b} - q_{0.15})), \quad (6)$$

where $q_{0.15}$ is the fifteenth percentile of the backbone output of that image. Using a percentile rather than a fixed level makes the gate work across scanners with different brightness. A second form of the gate, used where the selection of Section VIII-C chose it, is a soft step at the per image Otsu threshold followed by a maximum filter of fixed radius, so that the gate is identically zero beyond that radius from tissue.

The gated gain is applied multiplicatively to give the change $\mathbf{v}$, with the brightening part smoothed before it is added so that it cannot create a new high frequency edge,

$$\mathbf{v} = (\text{smooth}(\text{ReLU}(\alpha) \odot \mathbf{b}) + (\alpha - \text{ReLU}(\alpha)) \odot \mathbf{b}) \odot \mathbf{g}, \quad (7)$$

where smooth is a three by three mean filter. The darkening part is left unsmoothed, so the brightening is spread over a neighbourhood while a darkening acts where it was proposed. In the background, defined by the complement of the gate eroded by two pixels, a masked three by three mean filter is blended in at half strength, which lowers the background standard deviation without

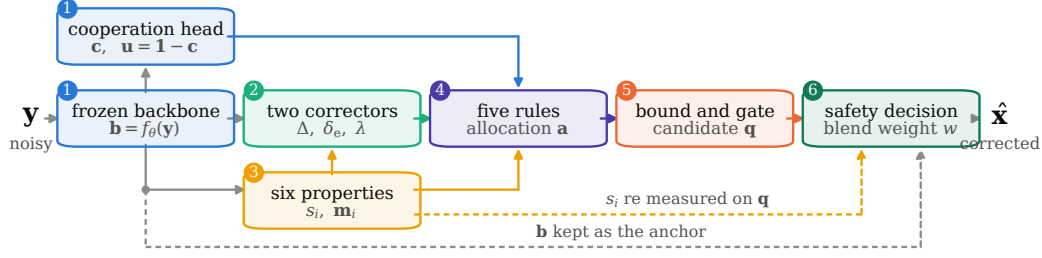


FIGURE 1. The six stages. A frozen backbone and a small head produce the image and the cooperation map. Two networks propose changes. Six properties are measured. Five rules allocate. The change is bounded and gated. Three constraints decide how much is kept.

letting blur cross into tissue. Finally the edge proposal is added and the result is clipped to the valid range,

$$\mathbf{q} = \text{clamp}(\mathbf{b} + \mathbf{v} + \delta_e \odot \mathbf{g}, 0, 1). \quad (8)$$

B. Stage six, the safety decision

The candidate $\mathbf{q}$ is not returned directly. Three constraints are checked on it.

Aggregate quality must not fall. Define for any image $\mathbf{t}$ the energy

$$E(\mathbf{t}) = 1 - \frac{1}{6} \sum_{i=1}^6 s_i(\mathbf{t}) + 0.1 \sum_{i=1}^6 \max(0, 0.5 - s_i(\mathbf{t})), \quad (9)$$

which is small when the image is good, and whose second term penalises any single property below one half, so a good average cannot hide one broken property. The first constraint requires $E(\mathbf{q}) \leq E(\mathbf{b}) + 0.10$.

No property may be sacrificed for another. The second constraint requires that at least one property improves by more than 0.01 and that no property falls by more than 0.10.

The image must not move far. The third constraint requires $\|\mathbf{q} - \mathbf{b}\|_2 / \|\mathbf{b}\|_2 < 0.2$.

Let $n \in \{0, 1, 2, 3\}$ be the number of constraints that hold. The output is

$$\hat{\mathbf{x}} = (1 - w) \mathbf{b} + w \mathbf{q}, \quad w = \begin{cases} 1, & n \geq 2, \\ 0.05 + 0.10 \frac{n}{3}, & n < 2. \end{cases} \quad (10)$$

Otherwise the output falls back to the backbone while retaining five percent when none holds and about eight when one does, so the decision remains visible rather than silently discarded. The candidate is accepted in full whenever any two constraints hold, so one violated constraint does not by itself withhold the change. This is a majority test on three measured conditions and not an enforcement of three bounds, and Section IX-H reports how often each holds.

VII. Stability and Guarantees

Section IX-G tests both results of this section on real images.

A. What stability means here

This is not stability in the sense of Lyapunov and we claim no fixed point. The quantity that must not behave erratically is the decision of the rule layer, a function of the five rule truth values, which are exactly what shift when the backbone or the scanner changes.

Definition 1 (Stability of the allocation). Let $\mathbf{r} = (r_1, \dots, r_5) \in [0, 1]^5$ be the vector of rule truth values at one pixel and let $a(\mathbf{r})$ be the allocation of Equation (4) at that pixel with the parameters held fixed. The allocation is said to be stable with constant L if

$$|a(\mathbf{r}) - a(\mathbf{r}')| \leq L \|\mathbf{r} - \mathbf{r}'\|_\infty \quad \text{for all } \mathbf{r}, \mathbf{r}' \in [0, 1]^5. \quad (11)$$

In words, L is the worst case ratio between a change in the decision and the change in the evidence that caused it. Without such a bound the layer could swing from no correction to all of it because one property score moved slightly, which is the behaviour a clinical user cannot audit.

B. The rule layer is stable, with an exact constant

Theorem 1 (Allocation stability). The allocation of Equation (4) satisfies Definition 1 with

$$L = C_1 w_1 + C_2 w_2 + C_3 w_3 + C_4 w_4 + C_5 w_5, \quad (12)$$

where $w_k = \sigma(\theta_k)$ are the learned rule weights. The constant is attained, so it is the exact Lipschitz constant and not merely an upper bound. Moreover $L < C_1 + C_2 + C_3 + C_4 + C_5 = 0.75$ for every possible setting of the parameters.

Proof:

The allocation is affine in $\mathbf{r}$ with coefficients $\pm C_k w_k$, so subtracting two values and applying the triangle inequality gives $|a(\mathbf{r}) - a(\mathbf{r}')| \leq \sum_k C_k w_k |r_k - r'_k| \leq L \|\mathbf{r} - \mathbf{r}'\|_\infty$. The bound is attained when each $r_k - r'_k$ takes the sign of its coefficient, and $L < 0.75$ because every C_k is positive and every w_k lies strictly inside $(0, 1)$. ■

Corollary 1 (Behaviour under a change of backbone). Suppose replacing one backbone by another changes the rule truth values at a pixel by at most ϵ in the maximum norm, with the parameters unchanged. Then

TABLE 2. The six clinical properties in closed form. Here e is the Sobel edge magnitude of $\mathbf{b}$, $\bar{e}$ its mean, $b_e = \mathbf{1}[e > \bar{e}]$, σ_k and Var_k the local standard deviation and variance over a $k \times k$ window, $\bar{\sigma}$ the mean of σ_9 , $v(\cdot)$ the depth profile averaged across width, ∇_z the derivative along depth, ρ_h the one pixel horizontal autocorrelation, sigm the logistic function, ε a small constant, and cv the local coefficient of variation of the log residual. The named subterms are expanded in the supplementary file.

P_i	Property	Score s_i	Map $\mathbf{m}_i$	Where the map is large	τ_i
P_1	edge continuity	$0.4 c_{\text{cont}} + 0.3 c_{\text{efr}} + 0.3 c_{\text{ang}}$	$(1 - \bar{e})(1 - b_e)$, $\bar{e} = e/(\max e + \varepsilon)$	off detected edges, where a boundary should continue but does not	0.60
P_2	local contrast	$0.20 s_{\text{range}} + 0.35 s_{\text{cnr}} + 0.20 s_{\text{var}} + 0.25 s_{\text{cov}}$	$[\text{ReLU}(\bar{\sigma} - \sigma_9(\mathbf{b})) / (\bar{\sigma} + \varepsilon) + 0.3 \mathbf{1}[\sigma_9(\mathbf{b}) < 0.01]]_{[0,1]}$	where σ_9 falls below the image mean, and in nearly flat tissue	0.50
P_3	flat smoothness	$0.6 q_{\text{var}} + 0.4 q_{\text{res}}$	$F \odot \sigma_5(\mathbf{b}) / (\max \sigma_5(\mathbf{b}) + \varepsilon)$	inside flat regions that still carry residual variation	0.75
P_4	structural coherence	$0.4 s_{\text{self}} + 0.3 s_{\text{reg}} + 0.3 s_{\text{vert}}$	$e(e) / (\max e(e) + \varepsilon)$	where the edge map is itself rough, a sign of broken structure	0.50
P_5	speckle conformity	$0.7 \text{mean}(\exp(- cv - cv^* /0.15)) + 0.3 \exp(-5 \rho_h(z))$	$1 - \exp(- cv - cv^* /0.15)$	where cv departs from the value expected for OCT speckle	0.30
P_6	layer visibility	$0.4 s_{\text{clr}} + 0.3 s_{\text{dep}} + 0.3 s_{\text{smo}}$	$\text{Var}_7(e) / (\max \text{Var}_7(e) + \varepsilon)$	where the local variance of the edge map is high across depth	0.70

the allocation at that pixel changes by at most $L\varepsilon$, with L from Equation (12).

This concerns the routing rule alone, not whether image quality transfers, which is the empirical question of Section IX-C.

C. When the correction cannot reduce contrast

Write $\text{CNR}(\mathbf{t}) = (\mu_{\mathcal{T}}(\mathbf{t}) - \mu_{\mathcal{B}}(\mathbf{t})) / \sigma_{\mathcal{B}}(\mathbf{t})$ for the separation between tissue and background divided by the variability of the background.

Theorem 2 (Contrast margin). Let $\eta_{\mathcal{T}} = \mu_{\mathcal{T}}(\mathbf{q}) - \mu_{\mathcal{T}}(\mathbf{b}) \geq 0$ be the gain in tissue mean and let $\eta_{\mathcal{B}} \geq \mu_{\mathcal{B}}(\mathbf{q}) - \mu_{\mathcal{B}}(\mathbf{b})$ bound the leakage into the background. Suppose the tissue and background sets are the same for $\mathbf{b}$, $\mathbf{q}$ and $\hat{\mathbf{x}}$, that $\sigma_{\mathcal{B}}(\mathbf{b}) > 0$ and the correction does not raise the background variability, so $\sigma_{\mathcal{B}}(\mathbf{q}) \leq \sigma_{\mathcal{B}}(\mathbf{b})$, and that the bounded numerator $(\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta_{\mathcal{B}})$ is non negative, which holds whenever the backbone already separates tissue from background and $\eta_{\mathcal{T}} \geq \eta_{\mathcal{B}}$. Then the output of Equation (10) satisfies

$$\text{CNR}(\hat{\mathbf{x}}) \geq \frac{(\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta_{\mathcal{B}})}{\sigma_{\mathcal{B}}(\mathbf{b})}. \quad (13)$$

In particular, whenever $\eta_{\mathcal{T}} > \eta_{\mathcal{B}}$ the correction cannot reduce the contrast to noise ratio.

Proof:

The mean over a fixed pixel set is linear, so $\mu_{\mathcal{T}}(\hat{\mathbf{x}}) = \mu_{\mathcal{T}}(\mathbf{b}) + w\eta_{\mathcal{T}}$ and $\mu_{\mathcal{B}}(\hat{\mathbf{x}}) \leq \mu_{\mathcal{B}}(\mathbf{b}) + w\eta_{\mathcal{B}}$, which bounds the numerator of $\text{CNR}(\hat{\mathbf{x}})$ below by the numerator of Equation (13). On the background $\hat{\mathbf{x}}$ is the same convex combination pixel by pixel, so Minkowski's inequality on the centred values gives $\sigma_{\mathcal{B}}(\hat{\mathbf{x}}) \leq (1 - w)\sigma_{\mathcal{B}}(\mathbf{b}) + w\sigma_{\mathcal{B}}(\mathbf{q}) \leq \sigma_{\mathcal{B}}(\mathbf{b})$. Dividing the two bounds gives the claim, and $w > 0$ in every branch of Equation (10). ■

Three parts of the design make the hypotheses of Theorem 2 hold in practice. The gate of Equation (6) keeps $\eta_{\mathcal{B}}$ small by suppressing change in the dark region, the brightness branch of the gain network cannot be negative, which keeps $\eta_{\mathcal{T}}$ non negative, and the

background smoothing lowers $\sigma_{\mathcal{B}}(\mathbf{q})$ rather than raising it.

Both results are deliberately narrow. Theorem 1 bounds how far the routing decision can move, not whether it is correct, Theorem 2 concerns contrast alone and says nothing about the other five measures, and neither promises that the output is closer to the clean reference, which Section IX-B shows it usually is not. The guarantees concern the wrapper, not the denoiser underneath it.

VIII. Training

A. The six loss terms

Let $\mathbf{x}$ be the clean reference, $\mathbf{b}$ the backbone output and $\hat{\mathbf{x}}$ the wrapper output. The objective of Equation (1) has the following terms.

Fidelity. Let $\Delta = \text{PSNR}(\mathbf{b}) - \text{PSNR}(\hat{\mathbf{x}})$ be the drop against the backbone in decibels and Δ_s the corresponding drop in the structural similarity index [22]. Neither is charged until it exceeds a dead zone,

$$\mathcal{L}_{\text{fid}} = 5[\Delta - d]_+ + 15[\Delta - d - 0.4]_+^2 + 20[\Delta_s - 0.005]_+, \quad (14)$$

where $[\cdot]_+$ is the positive part. The term is flat while the drop stays below d decibels, linear for the next four tenths of a decibel and quadratic beyond that, so a small loss of fidelity is free and a large one is expensive. The dead zone d is one of the two quantities selected per backbone in Section VIII-C.

Clinical improvement. This term charges the wrapper only when a measure falls short of a target taken from the backbone, so a gain in one measure cannot be bought by a collapse in another. Let q_m be the value of quantity m on the wrapper output and $t_m = \kappa_m q_m(\mathbf{b})$ the target, the backbone value of the same quantity scaled by a fixed factor. Writing $\varepsilon_m = t_m - q_m$,

$$\mathcal{L}_{\text{clin}} = \frac{1}{\sum_m w_m} \sum_{m \in M} w_m \left(3[\varepsilon_m]_+ + [\varepsilon_m]_+^2 + \frac{1}{2} s_m \varepsilon_m^2 \right). \quad (15)$$

The set M holds nine quantities, the three preservation scores of local contrast, boundary gradient and texture,

then EPI, CNR, BS, ENL, SNR and the ratio of the backbone squared error to the wrapper squared error. Their weights w_m are 1, 5, 1, 15, 40, 5, 8, 40 and 10 and their factors κ_m are 1.12, 1.12, 1.00, 1.05, 1.20, 1.12, 1.10, 1.15 and 0.87, the last permitting about six tenths of a decibel. The indicator s_m is one for ENL, SNR and the squared error ratio, a symmetric pull that holds those three near target instead of letting them run. None of the nine is used at test time.

Preservation scores. This term acts on four scores g_j separate from the six properties of Table 2, comparing the corrected output with the backbone output on local contrast, the gradient across layer boundaries, the local variance carrying texture, and the gradient magnitude at edges. With $\delta_j = g_j(\hat{\mathbf{x}}) - g_j(\mathbf{b})$, the backbone value fixed,

$$\mathcal{L}_{\text{pred}} = \sum_{j=1}^4 v_j \left(3[-\delta_j]_+ - 2.5[\delta_j]_+ + 2\left[\frac{1}{2} - g_j(\hat{\mathbf{x}})\right]_+ \right), \quad (16)$$

where $[\cdot]_+$ is the positive part and v_j is 3, 2, 2 and 1 in that order. A drop costs more than a rise earns, and any score below one half is charged again. The thresholds τ_i of the six properties play no part here. They enter Equation (3), but the rule layer that evaluates it runs without gradient tracking, so no path reaches them and all six hold their Table 2 values throughout training.

Cooperation. The wrapper should act where the backbone leaves room and stay quiet elsewhere. We take the Pearson correlation ρ between the demand map $\mathbf{u}$ and $\boldsymbol{\pi}$, the summed proposal strength the rule layer reads, and penalise disagreement,

$$\mathcal{L}_{\text{coop}} = \frac{1}{2} \text{ReLU}(1 - \rho(\mathbf{u}, \boldsymbol{\pi})). \quad (17)$$

The Pearson correlation is differentiable everywhere, invariant to an affine change of either map, and introduces no window size to tune.

Edge supervision. The additive branch is trained to reproduce the gradient that the backbone lost, measured with the Sobel operator ∇ and weighted toward the places where the clean reference has an edge. Let $\mathbf{w}$ be the clean Sobel magnitude divided by its mean and clipped at ten, $\tilde{\mathbf{w}}$ its square normalised again to unit mean, and $\tilde{\mathbf{g}}$ a Sobel magnitude divided by its own maximum. With $\delta = [1.05 \text{EPI}(\mathbf{b}) - \text{EPI}(\hat{\mathbf{x}})]_+$,

$$\begin{aligned} \mathcal{L}_{\text{edge}} = & 10 \|\sqrt{\tilde{\mathbf{w}}} \odot (\nabla \hat{\mathbf{x}} - \nabla \mathbf{x})\|_2^2 \\ & + 10 \|\sqrt{\tilde{\mathbf{w}}} \odot (\tilde{\mathbf{g}}(\hat{\mathbf{x}}) - \tilde{\mathbf{g}}(\mathbf{x}))\|_2^2 + 45(3\delta + \delta^2). \end{aligned} \quad (18)$$

The first part is the gradient the correction must restore. The second compares normalised edge magnitude, which is insensitive to overall brightness. The third refuses an edge preservation index below five percent above the backbone.

Bounds and background protection. The last term collects the constants that keep the change small, smooth

and confined. Let a be the mean absolute change over the upper half of the image, e the mean absolute output of the additive edge branch, σ_B the standard deviation over the lower quarter, μ_T the mean over the upper half and $\mathbf{m}$ a soft mask on intensities below the fifteenth percentile of the backbone output,

$$\begin{aligned} \mathcal{L}_{\text{reg}} = & 40[0.002 - a]_+ + 200[a - 0.006]_+^2 + 2\|\nabla \boldsymbol{\alpha}\| \\ & + 30[e - 0.06]_+^2 + 80[\sigma_B(\hat{\mathbf{x}}) - 0.75\sigma_B(\mathbf{b})]_+ \\ & + 100\left[\frac{\sigma_B(\hat{\mathbf{x}})}{\sigma_B(\mathbf{b})} - 0.95\right]_+ + 150\left[0.99 - \frac{\mu_T(\hat{\mathbf{x}})}{\mu_T(\mathbf{b})}\right]_+ \\ & + 2\|\hat{\mathbf{x}} - \mathbf{b}\| \odot \mathbf{m}, \end{aligned} \quad (19)$$

where a bar denotes the mean over pixels and the first two parts are clipped at five. The floor stops the wrapper from learning to do nothing and the ceiling from overshooting fidelity, the third keeps the allocation map smooth so that it leaves no Sobel trace of its own, the fourth bounds the edge branch, and the last three hold the background noise and the tissue brightness, the denominator and the numerator of the signal to noise ratio.

B. Data and optimisation schedule

Training uses paired data, a noisy scan and a clean reference averaged over repeated acquisitions, and the reference enters the fidelity and clinical terms and supervises the edge branch. Testing uses no clean image at any point, since the six properties read only the noisy input and the backbone output, and the reference only scores the results. The backbone is trained first, on the training split with the squared error alone, and is then frozen. The wrapper is then trained in two stages. Only the clinical terms are active at first, so the correctors learn a useful direction, and the fidelity term switches on at the midpoint, so they learn how far they may go. Both stages use AdamW with weight decay 10^{-4} on 96×96 crops in batches of sixteen, eight for DnCNN and SwinIR, for ten epochs in all, with base learning rates of 2×10^{-4} for the corrector networks, 5×10^{-4} for the cooperation head and the strength predictor and 3×10^{-4} for the rule layer, and the output layers of the correctors and the background smoother at two to eight times the corrector rate. The supplementary file restates the procedure as pseudocode.

Throughout both stages the normalisation statistics of f are held fixed. DnCNN is the one backbone with batch normalisation, and left in training mode its buffers move during fitting, so the reference against which every clinical change of that backbone is measured would drift by about half a decibel and would change its boundary sharpness while the correction was being fitted against it.

C. How the operating point was chosen

The terms of Equation (1) carry fixed relative weights, the same for every backbone and seed. The clinical term dominates at 6, the property term is 1, the cooperation term is 0.2, and the fidelity term is 0 in the first stage and 1 in the second, which is what the two stage schedule means here, and the edge and regulariser terms are 1, since their internal coefficients already carry their scale. The parser also accepts ten weight flags, for the clinical term, the two contrast measures, fidelity in two forms, the edge term, three per measure preservations and efficiency. Not one of them reaches the loss. Two are printed at start up, which is the only thing that made any of them look active, and the release notes list all ten. Two quantities are selected per backbone on validation, the dead zone d of Equation (14) and the gate of Equation (6). The grid crosses the two gate rules with dead zones of 0.3, 0.6 and 0.9 dB and the tissue-bounded gate with radii of 41 and 61 pixels. It chose the tissue-bounded gate at radius 41 with a dead zone of 0.6 dB for NAFNet and the percentile gate at 0.9 dB for the other three. Everything else is held at one value throughout. Each setting is trained at three seeds and scored on validation. A setting is admissible only if it clears its bounds at its worst seed rather than on average, and among the admissible settings we take the one with the largest lower confidence bound on its weakest clinical measure. When no setting clears the first pair of bounds we relax them one step at a time and record the step that was needed, so a backbone that required a looser condition is visible rather than hidden. Every source of randomness is seeded, up to the nondeterminism of the convolution kernels, and the checkpoints chosen on validation are the checkpoints scored on test.

D. Initialisation and the choice of every constant

Every constant appears with its value where it acts and has one of four origins. Fixed structural choices, never tuned, are the sharpness $\kappa = 10$ of Equation (3), the steepness 20 of Equation (6), the window sizes, the quarter resolution of the gain network, the nominal bound ± 0.08 of the edge branch, the thresholds τ_i of Table 2, taken from the clinical literature and unchanged by training to four decimals in every run, and the budgets B and C , tied by the identity of Section V-B. Set once on the validation split while the method was built and never revisited are $\alpha_{\max} = 0.15$, the gain bound ± 0.20 , the gate percentile 0.15, the three tolerances of Section VI-B, the half strength background blend and the learning rates. Selected in the grid of Section VIII-C, on the validation split only, are the dead zone d and the gate rule. Learned by the objective are β , θ_1 to θ_5 , τ_0 , β_0 and the scale of the edge branch. Every layer uses its default initialisation with one deliberate exception, the bias of the brightness channel of the gain network starts

at 0.5, so the raw gain begins at $0.5\alpha_{\max}$ in flat tissue rather than on a flat region of the loss. Nothing was ever chosen on the test split, and Section IX-D reports how much the results move when the constants of the rule layer are changed, the direct test of whether any of them materially affects the result.

IX. Tests and Results

A. Setup

Data. PKU37 [12] provides 37 cross sections of 640×640 pixels from a spectral domain scanner, references averaged over 50 frames, split by clean image identity into 27, 5 and 5, giving 1388, 173 and 173 pairs, and is the only training domain. Duke17 [13] provides 17 paired B-scans of 970×520 acquired at another institution, of which we use the 16 stored under the naming the released loader reads, and Duke2013 [14] the 18 volumes of 900×450 of its test split. Both Duke sets come from the same manufacturer and the 2013 evaluation reuses some 2012 material, so we call them two external datasets rather than two independent scanners.

Transfer protocol. Every model is trained once on PKU37 and applied to the Duke sets directly, with nothing fitted, recalibrated or selected on Duke data beyond the per image quantity the gate of Equation (6) computes by the same rule as on PKU37. The original submission instead adapted the wrapper for fifteen further epochs on the other subjects of the target set, leave one subject out, backbone frozen, with an L2 penalty holding every trainable weight near its PKU37 value. We rerun that protocol for NAFNet and report it only in the supplementary file, for the reason given in Section IX-C.

Measures. Peak signal to noise ratio against the clean reference measures fidelity. Six clinical measures, given as percentage change from backbone to corrected output, are contrast to noise ratio for tissue against background, tissue contrast index for dynamic range in tissue, edge preservation index for agreement with the reference edge map, boundary sharpness for the gradient across layer boundaries, equivalent number of looks for speckle suppression, and signal to noise ratio against the noise floor. Each is computed per image then averaged, never pooled, since pooling inflates the result. The 173 pairs come from five clean image identities, so every count and test below describes behaviour across repeated acquisitions of five anatomies and not across patients.

B. In distribution results

Table 3 reports the PKU37 test set and Fig. 2 the visual comparison across the four backbones. The wrapper raises contrast on every backbone, by 3.5 to 11.3 percent, and the speckle measure by 23.7 to 57.1 percent, for under a decibel of fidelity in the worst case and 0.17 dB in the best. Twenty one of the 24 backbone by measure pairs improve by more than one seed spread, one improves

TABLE 3. PKU37 test set of 173 images, then Duke17 and Duke2013 with no adaptation of any kind. Columns give the backbone peak signal to noise ratio in decibels, its change after correction with the spread across three seeds, and the six clinical measures as percentage change from backbone output to corrected output, bold where the mean exceeds the seed spread. The component removal block and the alternative operator block below use the selected NAFNet checkpoint on the same 173 test images, with one component removed or one operator substituted. Full spreads are in the supplementary file.

Backbone	Backbone PSNR (dB)	Δ PSNR (dB)	Δ CNR	Δ TCI	Δ EPI	Δ BS	Δ ENL	Δ SNR
PKU37 test set, 173 images								
NAFNet	30.75	-0.18 $\pm$ 0.02	+3.49	+1.91	+1.20	+3.89	+23.74	+11.10
DnCNN	30.68	-0.98 $\pm$ 0.21	+11.30	+0.99	+4.14	-1.44	+57.14	+22.95
SwinIR	29.87	-0.17 $\pm$ 0.06	+7.33	+11.45	+2.68	+1.82	+28.32	+11.54
KBNet	32.04	-0.90 $\pm$ 0.08	+6.93	+5.92	+0.00	+2.58	+25.30	+7.95
Duke17, no adaptation								
NAFNet	27.03	-0.17 $\pm$ 0.01	+7.51	+4.00	+1.37	+4.15	+21.35	+5.43
DnCNN	26.89	-0.29 $\pm$ 0.05	+12.19	-1.47	+3.49	-2.24	+39.42	+13.55
SwinIR	27.06	-0.19 $\pm$ 0.05	+8.68	+13.26	+2.69	+1.90	+29.46	+11.60
KBNet	27.59	-0.51 $\pm$ 0.05	+5.34	+7.19	-0.50	+1.61	+7.49	-1.81
Duke2013, no adaptation								
NAFNet	27.33	-0.35 $\pm$ 0.01	+7.37	+1.41	+0.85	+0.51	+28.80	+9.29
DnCNN	27.21	-0.42 $\pm$ 0.05	+12.73	-2.21	+2.99	-5.85	+49.91	+17.59
SwinIR	26.89	-0.19 $\pm$ 0.06	+9.26	+11.86	+2.55	+2.30	+30.90	+13.31
KBNet	28.12	-0.59 $\pm$ 0.04	+7.16	+4.83	-0.35	+3.06	+8.70	-0.90
NAFNet on PKU37, one component removed								
Full method		-0.17	+3.71	+2.42	+1.21	+4.16	+23.34	+10.74
No rule layer		-0.38	+0.83	+0.38	+0.99	+4.27	+25.45	+10.44
No edge branch		-0.37	-2.71	-6.54	+0.55	+0.65	+20.28	+7.30
No cooperation map		-0.15	+2.51	+1.15	+1.10	+3.87	+23.84	+10.61
No background smoothing		-0.18	+3.03	+6.88	+0.92	+3.99	+5.75	+2.85
NAFNet on PKU37, alternatives of similar complexity on the same backbone output								
Uniform allocation, trained		-0.42	+4.11	+11.58	+0.71	+3.83	+16.18	+5.80
Unsharp masking		-0.67	+0.63	+39.45	-3.59	+25.13	-35.31	-20.57
Adaptive equalisation		-4.43	+6.48	+53.03	-3.45	+24.62	-38.60	-25.13

inside it, one is unchanged and one falls, the boundary sharpness of DnCNN. The gain holds image by image, not only in the mean. Contrast rose on all but twelve of the 2076 image and seed combinations, none of the twelve by as much as half a percent. It is largest where the backbone leaves the most to repair, the per image gain correlating negatively with backbone fidelity on all four.

C. Transfer to other datasets

Table 3 reports the no adaptation protocol on Duke17 and Duke2013. All six clinical measures are positive on

both target sets for NAFNet and SwinIR, the fidelity cost stays inside 0.35 dB for those two, and 40 of the 48 backbone by measure by set pairs are positive, the eight exceptions being the tissue contrast index and the boundary sharpness of DnCNN and the edge preservation and the signal to noise ratio of KBNet. Contrast rises on every backbone and both sets.

The adapted protocol of the original submission is not reported here. Each of its folds keeps the epoch that scores best on the subject it then reports, so the numbers are selected on the data they describe and are not an estimate of what adaptation achieves. We rerun it for

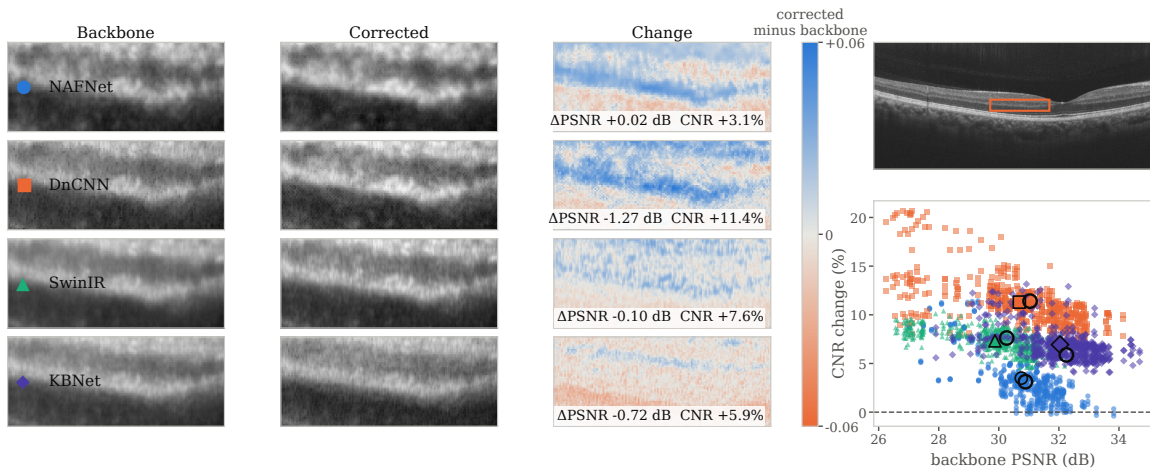


FIGURE 2. Visual comparison on the PKU37 test image of median backbone fidelity, one row per backbone, in the 56 by 136 pixel band marked on the thumbnail where the NAFNet correction was largest. The eight gray cells share one display range, set to the values they contain so that nothing is clipped, and the signed difference uses a fixed scale of plus or minus 0.06. The mean absolute change is about one percent of the intensity range, most for SwinIR and least for KBNet, from the seed 0 checkpoints of Table 3. The scatter places this image, ringed, among all 173 test images at three seeds for four backbones, with the twelve NAFNet points below zero marked. Full resolution pairs for every backbone are in the supplementary file.

the record and give it in the supplementary file as a historical diagnostic, clearly labelled. The frozen model is the one this paper reports throughout.

D. Which properties and which constants matter

One property at a time is switched off at test time, by setting its score to the satisfied value and its failure map to zero, and the whole test set is re evaluated. Only P_5 , P_1 and P_2 move any measure by more than a quarter point. P_5 costs the most at 3.19 points of the speckle gain, P_1 1.72 and P_2 0.74, while P_3 , P_4 and P_6 change nothing by more than 0.23 points. We keep all six, because they are the vocabulary of the rule layer and of the explanation shown to the reader, and one that rarely fires here can fire on another scanner.

The rule layer holds a base level, five rule weights and five budgets. The budgets are tied by the identity of Section V-B, so they stay fixed. We vary the base level and the two rule weights that carry the allocation, and replace the Lukasiewicz conjunction by the product and by the minimum. No variation turns any clinical measure negative. Across the fourteen the contrast gain lies between 3.5 and 4.5 percent, the tissue contrast index between 2.3 and 2.9, the speckle measure between 22.6 and 23.5 and the fidelity cost between 0.16 and 0.23 dB. Only the base level has a visible effect, lowering it raising contrast and costing fidelity together, while the two weights move the result by under a fifth of a point. The product and the minimum take contrast from 3.7 to 3.5 percent and leave the rest within one part in fifty, so the result rests on the shape of the rule layer and not the particular conjunction. No setting of either gate recovers boundary sharpness on DnCNN.

E. What each component contributes, and how it compares

The component block of Table 3 removes one component at a time. Replacing the rule layer by a uniform allocation cuts the contrast gain from 3.7 to 0.8 percent and the tissue contrast index from 2.4 to 0.4 percent while doubling the fidelity cost, the clearest evidence that the allocation and not the capacity of the correctors produces the clinical gain. Removing the edge branch turns both measures negative, and removing the cooperation map costs about a third of the contrast gain. Removing background smoothing leaves contrast, edge preservation and boundary sharpness almost intact while the speckle measure collapses from 23.3 to 5.8 percent and the signal to noise ratio from 10.7 to 2.9 percent.

Three alternatives act on the same frozen backbone output. The first is the same architecture, budget, objective and ten epochs, with the rule layer replaced by a uniform allocation throughout training rather than only at test time. It was fitted before the gate search, so it carries the percentile gate and not the tissue-bounded gate NAFNet was selected with, which makes

it matched in architecture and not in configuration. The other two are classical operators, unsharp masking and contrast limited adaptive histogram equalisation, tuned on the validation split. Uniform allocation reaches a larger contrast gain and a much larger tissue contrast index than the full method, for 0.42 dB against 0.17 dB, and is worse on four of the six measures, so the two are not ordered. What the full method buys with its fidelity is the account of why each change was made. Both classical operators raise the tissue contrast index far above anything the wrapper reaches, and adaptive equalisation also raises contrast above it while unsharp masking does not. Both fail elsewhere, lowering edge preservation by about 3.5 percent and the speckle measure by about 36 percent while raising boundary sharpness by about 25 percent by sharpening noise, and adaptive equalisation costs 4.4 dB.

F. Does the cooperation map predict error

Called a confidence map in the original submission, this component was tested against the absolute error of the backbone output over 173 images, and it does not predict that error. The quantity tested is the demand map $\mathbf{u} = \mathbf{1} - \mathbf{c}$, which is high where the wrapper is told to act, so a demand map that tracked backbone error would give a positive rank correlation. The measured value is -0.258, ordered the wrong way rather than merely uninformative, and removing the pixels it ranks as least reliable raises the remaining error to 1.88 times the starting value where an oracle ranking reaches 0.07, a sparsification error of 0.776. The map is nevertheless useful, since replacing it by a constant lowers the contrast gain from +3.71 to +2.51 percent and worsens five of the six clinical measures. Nothing in the objective shows the head the error of the backbone, so we call it a cooperation map rather than a confidence.

G. Do the theoretical conditions hold on real data

We perturb the rule truth values and compare the ratio of Equation (11) with the constant of Equation (12), and we measure the tissue gain, the background leakage and the ratio of background standard deviations on every image against the hypotheses of Theorem 2. The stability bound holds with room to spare, a constant of 0.5489 from Equation (12) against a largest measured ratio of 0.4679 over 173 images. The constant is exact for the allocation as a function of the rule truth values, so the gap is not slack in the proof. It says the perturbations a real image produces never reach the worst case the constant allows. The allocation map spans 0.556 to 0.793 with mean 0.726 and standard deviation 0.050, and 0.00 percent of pixels sit at either end of its clamp, direct evidence that the rule layer does work after the repair of Section V-B.

The margin condition holds on 139 of 173 images and the ratio condition on 94, but both hold together on only 60, and the theorem needs both. Where a hypothesis fails the theorem predicts nothing, and the measured contrast change there does not differ systematically in sign, so the conditions are sufficient and not necessary. Both leakage terms of Theorem 2 sit far below the mean absolute change over the image, 0.0067, at -0.0007 in tissue and -0.0017 in background, with means near zero and a long negative tail, so the per image sign is what the theorem needs. Tissue is positive on 137 of 173 images and background negative on 107, the direction the decomposition assumes.

H. Safety, and whether the correction invents structure

A correction that raises contrast could sharpen a boundary that is not there or erase a faint one that is, and the clean reference lets us measure both. Two measures move against it. An edge appears where the reference has none on 0.48 percent of pixels after correction against 0.45 percent for the backbone output, a rise on 127 of 173 images with $p = 1.9 \times 10^{-12}$ under a two sided Wilcoxon signed rank test on the paired per image values, and the retention of structures whose reference contrast falls in the lowest decile falls from 40.5 to 37.6 percent, on 168 of 173 images with $p = 6.2 \times 10^{-30}$. The first is 0.03 points of invented edge rate, the second 2.9 points of retention, or 7.2 percent of where it starts. Neither is zero by construction, since the clamp bounds how far a pixel may move and not how many cross an edge threshold, and the discussion lists both. What separates it from a sharpening filter is Section IX-E, where the classical sharpeners lose edge preservation and the speckle measure while the wrapper gains on both. The two layer boundaries barely move. The inner limiting membrane shift falls from 4.73 to 4.38 pixels and the retinal pigment epithelium shift goes from 9.93 to 10.03, which fails a two sided signed rank test. Neither establishes that anatomy is preserved. Dark fluid like regions occupy 1.6 percent of the image area with a mean absolute change of 0.0022 inside them against 0.0067 over the whole image, and the largest change on any single image anywhere in the test set is 0.0178, so the correction is quieter where a dark region carries the finding. Boundary sharpness is not worsened on 104 of 173 images, a count, since the per image worst case is what matters for trusting a scan. Of the three constraints of Section VI-B, energy holds on 173 of 173 images and bounded change on 173, and only the no sacrifice constraint ever fails, on 173 minus 135 of them. No image falls below two, so the candidate is returned in full everywhere and the layer never acted on this test set.

I. Cost of moving to a new backbone

A new backbone means refitting the whole wrapper, not only its head. The trainable count is 168 630 shared parameters plus a head whose width follows the feature tensor it reads, and the head is what varies, 4659 for KBNNet, 7263 for NAFNet, 85 908 for SwinIR and 234 273 for DnCNN, where the first convolution is wide. Against a frozen backbone of about seven million that is 2.3, 2.5, 3.7 and 5.8 percent, so the shared part is constant and the cost of attaching it follows how wide a tensor the backbone exposes. Every backbone is fitted on the same 1388 pairs for the same ten epochs, so only the wall clock and the seed spread vary. Fitting one seed of the selected setting on a single RTX 5070 Ti takes 2.0 minutes for NAFNet, 2.5 for KBNNet, 3.9 for DnCNN and 8.3 for SwinIR, following the cost of one backbone forward pass. The seed spread of the fidelity cost is 0.02 dB for NAFNet, 0.06 for SwinIR, 0.08 for KBNNet and 0.21 for DnCNN. The backbone with the widest spread is the only one whose clinical measure falls.

X. Discussion

A. Why fidelity falls, and when that is the right trade

Fidelity falls whenever the wrapper acts, and it has to. A denoiser trained on squared error already sits near a local optimum of that error, so any change that raises contrast or sharpens a boundary moves away from it. What matters is the size of the payment. It is under a decibel, smaller than the difference between two acquisitions of the same eye, and it buys the quantities a grader reads.

Where the gain lands is decided by what each backbone leaves undone rather than by how good it is overall. On boundary sharpness NAFNet starts furthest below the clean reference and gains most, while DnCNN starts above it and is the only backbone that falls. On edge preservation KBNNet starts highest and is the only one left unchanged. SwinIR is the instructive case, since windowed attention smooths hard inside a window, so it leaves the most room on contrast and pays the least fidelity of the four. A deployment that cannot afford the payment lowers the gain bound $\alpha_{\max}$, which attenuates the multiplicative branch, though only bypassing the wrapper returns the backbone output exactly.

B. Limitations

The correction is bounded by construction, so a backbone that has destroyed a structure cannot be made to recover it. Two safety measures move against it, the rate at which an edge appears where the reference has none and the retention of the weakest structures, both reported in Section IX-H. Neither cost is zero, and the criterion that chose the operating point scores six clinical measures and fidelity and sees neither. We measured what that blindness costs, on validation so that the test split is spent once. The runner up setting for NAFNet

keeps more weak structure than the selected one, 43.9 percent against 37.7, and invents fewer edges. A user who values faint detail above contrast should therefore score these two during selection. We did not reselect on this evidence, since the criterion was fixed before the test split was scored. The clinical measures used here are established proxies, not outcomes, and no clinician read the images, so the safety analysis of Section IX-H is quantitative evidence rather than clinical validation. Finally, the external datasets are small, sixteen and eighteen images, so the transfer results are evidence of behaviour across scanners rather than a precise estimate of the gain on any one device.

XI. Conclusion

We presented a post-correction layer for OCT denoising that is attached to a frozen denoiser rather than replacing it. It states six clinical properties in closed form, turns them into a per-pixel decision through five named rules, and limits how far the resulting change may move the image, so a change can be traced back to the property that motivated it and the constraint that allowed it to stand. Across four backbone families the same rules, budgets and loss weights raise the clinical measures at a fidelity cost below one decibel, and on three of the four no measure falls at all. The exception is boundary sharpness on DnCNN, which starts above the clean reference, where the method has no room to act. The measures are proxies, and the next step is a reader study asking whether the changes are the ones a grader would want.

REFERENCES

- [1] D. Huang et al., “Optical coherence tomography,” *Science*, vol. 254, no. 5035, pp. 1178–1181, 1991.
- [2] J. W. Goodman, *Speckle Phenomena in Optics: Theory and Applications*. Roberts & Company, 2007.
- [3] S. K. Devalla et al., “DRUNET: A dilated-residual U-Net deep learning network to segment optic nerve head tissues in optical coherence tomography images,” *Biomed. Opt. Express*, vol. 9, no. 7, pp. 3244–3265, 2018.
- [4] K. Zhang et al., “Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising,” *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3142–3155, 2017.
- [5] L. Chen et al., “Simple baselines for image restoration,” in *Proc. ECCV*, 2022, pp. 17–33.
- [6] J. Liang et al., “SwinIR: Image restoration using Swin Transformer,” in *Proc. ICCV Workshops*, 2021, pp. 1833–1844.
- [7] A. Krull, T.-O. Buchholz, and F. Jug, “Noise2Void—learning denoising from single noisy images,” in *Proc. CVPR*, 2019, pp. 2124–2132.
- [8] E. van Krieken, E. Acar, and F. van Harmelen, “Analyzing differentiable fuzzy logic operators,” *Artif. Intell.*, vol. 302, art. 103602, 2022.
- [9] Y. Zhang et al., “KBNNet: Kernel basis network for image restoration,” *Tech. Rep. arXiv:2303.02881*, 2023. [Online]. Available: <https://arxiv.org/abs/2303.02881>
- [10] A. Buades, B. Coll, and J.-M. Morel, “A non-local algorithm for image denoising,” in *Proc. CVPR*, vol. 2, 2005, pp. 60–65.
- [11] M. A. Mayer et al., “Wavelet denoising of multiframe optical coherence tomography data,” *Biomed. Opt. Express*, vol. 3, no. 3, pp. 572–589, 2012.
- [12] M. Geng et al., “Triplet cross-fusion learning for unpaired image denoising in optical coherence tomography,” *IEEE Trans. Med. Imag.*, vol. 41, no. 11, pp. 3357–3372, 2022.
- [13] L. Fang et al., “Sparsity based denoising of spectral domain optical coherence tomography images,” *Biomed. Opt. Express*, vol. 3, no. 5, pp. 927–942, 2012.
- [14] L. Fang et al., “Fast acquisition and reconstruction of optical coherence tomography images via sparse representation,” *IEEE Trans. Med. Imag.*, vol. 32, no. 11, pp. 2034–2049, 2013.
- [15] J. Wang et al., “Deep learning for quality assessment of retinal OCT images,” *Biomed. Opt. Express*, vol. 10, no. 12, pp. 6057–6072, 2019.
- [16] Y. Gal and Z. Ghahramani, “Dropout as a Bayesian approximation: Representing model uncertainty in deep learning,” in *Proc. ICML*, 2016, pp. 1050–1059.
- [17] A. Kendall, Y. Gal, and R. Cipolla, “Multi-task learning using uncertainty to weigh losses for scene geometry and semantics,” in *Proc. CVPR*, 2018, pp. 7482–7491.
- [18] B. Lakshminarayanan, A. Pritzel, and C. Blundell, “Simple and scalable predictive uncertainty estimation using deep ensembles,” in *Proc. NIPS*, 2017.
- [19] A. d’Avila Garcez and L. C. Lamb, “Neurosymbolic AI: The 3rd wave,” *Artif. Intell. Rev.*, vol. 56, no. 11, pp. 12387–12406, 2023.
- [20] D. M. Stein et al., “A new quality assessment parameter for optical coherence tomography,” *Br. J. Ophthalmol.*, vol. 90, no. 2, pp. 186–190, 2006.
- [21] P. Tewarie et al., “The OSCAR-IB consensus criteria for retinal OCT quality assessment,” *PLoS ONE*, vol. 7, no. 4, art. e34823, 2012.
- [22] Z. Wang et al., “Image quality assessment: From error visibility to structural similarity,” *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, 2004.
- [23] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *Proc. ICLR*, 2019.
- [24] R.-E. Precup et al., “Evolving fuzzy models of shape memory alloy wire actuators,” *Rom. J. Inf. Sci. Technol.*, vol. 24, no. 4, pp. 353–365, 2021.
- [25] Y. Fan et al., “Model-data-driven image reconstruction with neural networks for ultrasound computed tomography breast imaging,” *Neurocomputing*, vol. 467, pp. 10–21, 2022.
- [26] M. Casquilho, R. Galhano, and J. Buescu, “Effect of random variation of the initial values of ordinary differential equations and its web-based application to chemical kinetics,” *Rom. J. Inf. Sci. Technol.*, vol. 28, no. 3, pp. 286–298, 2025.
- [27] Z. Han et al., “Unifying neural learning and symbolic reasoning for spinal medical report generation,” *Med. Image Anal.*, vol. 67, art. 101872, 2021.
- [28] A. Nguembang Fadja et al., “Neural-symbolic ensemble learning for early-stage prediction of critical state of Covid-19 patients,” *Med. Biol. Eng. Comput.*, vol. 60, no. 12, pp. 3461–3474, 2022.
- [29] Q. Chen, L. Zeng, and C. Lin, “A deep network embedded with rough fuzzy discretization for OCT fundus image segmentation,” *Sci. Rep.*, vol. 13, art. 328, 2023.
- [30] T.-W. Kim and K.-C. Kwak, “Pixel-level fuzzy rule attention maps for interpretable MRI classification,” *Symmetry*, vol. 17, no. 12, art. 2187, 2025.
- [31] A. Özkan, Y.-H. Li, and T. Sikora, “Steered-mixture-of-experts regression for image denoising with multi-model inference,” in *Proc. EUSIPCO*, 2023, pp. 546–550.
- [32] A. Özkan et al., “Denoising OCT images using steered mixture of experts with multi-model inference,” in *Proc. SPIE*, vol. 12830, art. 1283009, 2024.
- [33] E. Zamfir et al., “Complexity experts are task-discriminative learners for any image restoration,” in *Proc. CVPR*, 2025, pp. 12753–12763.
- [34] M. H. Wang and S. Qin, “Explainable neuro-symbolic artificial intelligence for automated interpretation of corneal topography and early keratoconus detection,” *Front. Artif. Intell.*, vol. 9, art. 1713747, 2026.